

Chaos Language Algorithm (CLA)

Experiments, Results, Failure Diagnosis, and Repair Programme

ZeroBot / Chaos Language Algorithm and OmegaSim project records

Evidence cutoff: July 23, 2026

Abstract

This report consolidates the durable CLA evidence through July 23, 2026. It covers the pure-Python `chaoslang` implementation, low- and high-dimensional symbolization, synthetic grammar and complete-code tests, Mackey–Glass and Lorenz–96 calibration, geometric chaos-detector candidates, and the complete OmegaSim A6 CLA sweep. The central result is not that every part of CLA is broken. Exact reconstruction, deterministic replay, frozen prefix/suffix separation, compact model transmission, and shuffled-null rejection are generally sound. Low-cardinality symbolization repairs the obvious high-dimensional alphabet explosion. A separately calibrated nearest-neighbor divergence detector also discriminated the bounded positive and control battery. The unresolved failure is the bridge from a dynamical trajectory to a compact, generalizing grammar: model-description cost is often larger than the suffix savings, the search tends to return one- or two-production grammars with no categories, and complete CLA codes lose to unigram, Markov, or LZ78 controls on the chaotic fixtures. The report separates direct measurements from interpretation and proposes diagnostic experiments that can distinguish representation failure, search failure, and MDL accounting failure before further OmegaSim claims.

1 Executive diagnosis

Code URL: <https://github.com/bgoertzel-sing/chaos-language-algorithm>.

Current scientific status. CLA is a well-tested experimental infrastructure and a partially calibrated symbolic coder, not yet a validated “chaos grammar” detector. The decisive split is:

- **Engineering integrity mostly works:** exact reconstruction, canonical persistence, deterministic replay, prefix-only fitting, and null controls have repeatedly passed.
- **Representation partly works:** direct Cartesian symbolization fails in high dimension; fitted low-cardinality microstates restore recurrence and finite-order predictability. Tested linear VAMP embeddings did not beat matched PCA/raw controls.
- **Synthetic grammar competence is conditional:** indexed and compact transmission can win on some constant-length substitution sources and at long enough horizons, but does not generalize across all planted families.
- **Attractor grammar coding fails:** across scalar, ordinal, recurrence, and full-vector representations, compact CLA loses to simple complete-code controls.
- **Geometric Stage A can work:** nearest-neighbor divergence v2 passed its six-fixture battery. The subsequent prefix-quantized Stage-B CLA code lost badly to LZ78.
- **OmegaSim is negative under the frozen CLA:** all 120 planned role-specific A6 cells were evaluated with three seeds and matched appraisal/linear/shuffled controls; every grammar was trivial (one or two productions, zero categories).

Recommended decision. Do not spend effort on another identical OmegaSim A6 sweep. First instrument the inducer so that proposal coverage, accepted moves, data savings, model-cost increments, and rejection reasons are observable. Then run an oracle-decomposition battery that separately tests representability, search, parameter/model coding, and sample complexity. Only after a planted hierarchical grammar and a fresh chaotic fixture pass complete-code controls should CLA be used to prioritize OmegaSim regimes.

2 What CLA currently implements

The main pipeline is

$$x_{1:n} \rightarrow \text{state representation} \rightarrow \text{symbolizer} \rightarrow s_{1:n}$$

→ candidate mining → greedy grammar induction

→ model plus data code.

The **chaoslang** package began as a dependency-light, pure-Python, backend-neutral MVP. It contains fixed-grid and fitted microstate symbolization; n-gram/suffix-trie chunk mining; contextual category proposals; exact grammar expansion and reconstruction; approximate historical scores; later complete indexed and compact binary model transmissions; literal, unigram, Markov-1, Markov-2, and canonical LZ78 controls; frozen-prefix training; and deterministic experiment artifacts. The backend-neutral fact layer leaves a seam for a later Hyperon/MeTTa implementation but the reported measurements are from the Python system.

Measurement	What it establishes	What it does not establish
Exact reconstruction	The stored grammar/parse expands losslessly to the symbol stream	Predictive validity, semantic categories, or useful compression
Complete-code win	For the frozen fixture and code, model plus suffix is shorter than every named complete control	A universal detector or proof of chaos
Shuffled-null rejection	The frozen statistic/code does not prefer the exact named null	Robustness to every surrogate or confound
Held-out n-gram loss	Finite-order predictability of the symbol stream	Predictive likelihood of the induced CLA grammar
Stage-A divergence pass	Local neighbors separate under the frozen geometric test	A compact grammar, source-law identification, or semantic structure
OmegaSim matched-control win	A particular appraisal trace exceeds its linear and shuffled controls on frozen proxy metrics	Nontrivial grammar, strange attractor, or causal semantics

3 Chronological experiment programme

3.1 Baseline symbolic and high-dimensional tests

Low-dimensional 256-sample tests reconstructed exactly and produced small grammars. Under the original equal-width rectangular partition, alphabet size grew rapidly with dimension. The clearest high-dimensional smoke used 1,024 points of 20-dimensional Lorenz-96:

Metric	Value
Input symbols	1,024
Unique compound symbols	510
Accepted rules	8
Accepted categories	0
Approximate score	1,021.2
Fit time	2.554 s
Exact reconstruction	true

Observed. Half the positions were unique symbols and the score stayed close to one proxy unit per input symbol. **Inferred.** The primary failure occurred before grammar induction: a Cartesian product partition makes recurrence sparse in high dimension. This run cannot imply that Lorenz-96 lacks grammar. Provenance: experiment 20260709T192728Z-lorenz96-1024-dim20-suffix-trie.

3.2 R256 lifted Lorenz-63 and representation controls

A first Phase-1 test lifted Lorenz-63 into 256 dimensions, projected it to a small kinetic representation, and fitted low-cardinality microstates. Real-minus-shuffled proxy bits improved from 2.10 under compound M1 to 39.33, and held-out order-2 n-gram perplexity improved from 433.18 to 3.42. This showed that the symbolization seam could repair alphabet explosion, but it did not isolate the benefit of VAMP from clustering.

The preregistered three-seed follow-up evaluated 36 VAMP configurations plus controls:

Method	Mean proxy margin	Mean n-gram perplexity
deeptime VAMP (36 rows)	8.085	11.628
direct raw xyz k-means	12.200	3.697
dependency-free reference	15.167	4.151
raw compound M1	1.867	257.769

All VAMP and control configurations had positive shuffled-surrogate margins; no categories emerged. Positive proxy margin was therefore not specific to VAMP. In the leakage-free rank-conditioned follow-up, selected VAMP beat the best matched PCA lift on both held-out loss and surrogate margin in 0/3 trajectories at $k = 8$ and 0/3 at $k = 16$, versus the preregistered requirement of at least 2/3. VAMP held-out next-symbol loss was worse in every seed/alphabet stratum. Linear VAMP development was stopped.

Provenance: 20260712T200000Z-lorenz63-lift256-phase1, 20260712T200200Z-lorenz63-r256-deeptime-multiseed, and 20260713T153839Z-lorenz63-rank-conditioned-vamp-tica.

3.3 Frozen held-out and complete-code calibration

The project then replaced the historical approximate “bits” proxy with frozen held-out comparisons and explicit model transmission. The results exposed model cost as a dominant problem.

Run	Decision	Key quantitative result
Held-out v1	Fail	Aggregate CLA 65,976.621 bits versus unigram 6,588.443; about 32.5 kbits of model cost per fixture dominated.
Joint chunk/category diagnostic	Local positive	On a tiny constructed fixture, an abstract joint move scored 75 proxy bits versus 76 initially, showing the combined move can exist in the search space.
Generalized complete code	Reject	Joint move saved 6,232 data bits but added 6,296 model bits: 34,416 versus 34,352 total, a 64-bit loss.
Indexed complete code	Local positive	Same frozen joint state saved 1,672 data bits for 816 added model bits, a net 856-bit win. This validated the indexed transmission, not automatic search or attractor coding.
Held-out indexed v2	Fail	Aggregate indexed CLA 21,989.324 bits versus unigram 6,739.904; Markov-1 was 18,153.874.
Synthetic de Bruijn-5	Fail	Positive indexed CLA 4,600.700 bits versus unigram 2,608.721, a 1,991.979-bit loss; shuffled null not promoted.

Observed. More economical serialization substantially reduced overhead and could reverse a small constructed decision. It did not close the gap on the held-out attractor fixtures. **Inferred.** At least two separable questions remain: (1) whether the right grammar is proposed/found and (2) whether a good grammar is encoded with competitive finite-sample complexity. A single aggregate MDL score currently hides these mechanisms.

3.4 Synthetic source calibration

The synthetic programme tested planted symbolic structure without chaotic state reconstruction. It produced real, but narrow, successes.

Experiment	Gate	Main result
Sample complexity	Pass	A repeated source crossed at the frozen horizon; indexed model 4,712 bits plus 5.151 suffix bits beat Markov-2 by 3,798.715 bits; the null never crossed.
Family generalization	Fail	One planted positive passed, but Fibonacci word missed by 288.059 bits even though its suffix code was much shorter than Markov-2; model cost decided the result.
Two constant-length morphisms	Pass	Both 2-uniform positives beat controls; neither exact-marginal shuffled null was promoted. Rudin–Shapiro margin was only 171.466 bits.
Universal controls	Fail	Cantor lost even to LZ78. Paperfolding beat LZ78 but lost to unigram by 1,711.071 bits due to a 7,040-bit model.
Compact model	Fail	Compact transmission saved 464–648 bits versus JSON. Four-state 2-uniform won by 350.562 bits; complementary 3-uniform lost to unigram by 195.269 bits.
Phase robustness	Fail	Compact CLA lost the two required phases to LZ78 by 1,497 and 1,371 bits; nulls rejected.
Fixed-model amortization	Pass	Positive lost at 4,096 symbols but crossed at 16,384 and remained ahead at 65,536; shuffled source never crossed.

Observed. CLA can exploit some substitution-generated regularities, especially when a fixed model cost is amortized over a long suffix. It is not uniformly competitive across closely related planted families. **Inferred.** The algorithm has some genuine grammar-coding ability; the failure is better described as poor coverage/efficiency/generalization than total absence of capability.

3.5 Attractor representation and coding tests

Long-horizon amortization did not transfer from synthetic sources to the chaotic fixtures:

Representation	Positive fixtures	Result against strongest controls
Scalar fixed partition, 65,536 suffix	Mackey–Glass and Lorenz–96 x5	72,723.386 vs Markov-1 25,006.425; 103,647.803 vs Markov-1 43,859.342
Ordinal patterns	Mackey–Glass and Lorenz–96 x6	9,235.184 vs LZ78 2,872; 10,224.372 vs LZ78 3,928
Recurrence-lag projection	Mackey–Glass and Lorenz–96 x7	9,954.129 vs LZ78 5,008; 13,697.457 vs unigram 7,814.621
Full-vector recurrence	Mackey–Glass and full 8D Lorenz–96	9,933.113 vs LZ78 5,304; 5,522.398 vs LZ78 2,000

Every integrity gate passed in these runs; compact CLA beat the older JSON CLA continuity control, and shuffled nulls were not promoted. The positives still lost. Thus codec engineering improved CLA relative to itself without making it competitive with external codes.

Provenance: 20260718T131500Z-attractor-amortization-v1, 20260718T171500Z-ordinal-attractor-v1, 20260719T011500Z-recurrence-attractor-v1, and 20260719T071500Z-vector-recurrence-attractor-v1.

3.6 Geometric Stage-A detector search

Because grammar coding was not yet calibrated, a separate Stage-A programme asked only whether a geometric statistic could distinguish fresh chaotic fixtures from stable and shuffled controls.

Candidate	Gate	Error pattern
Recurrence determinism	Fail	Both positives detected, but stable Lorenz-96 was a false positive with DET=1.0.
Paired divergence	Fail	Mackey-Glass false negative (-0.0067585); stable Lorenz-96 false positive (0.0106748).
Correlation 0-1	Fail	Both positives false negatives; stable controls correctly rejected.
Renormalized separation	Fail	Lorenz-96 positive detected; Mackey-Glass false negative (-0.0009623).
Conditional entropy v1	Protocol failure	Runner requested a nonexistent result attribute; no score was produced.
Conditional entropy v2	Fail	Both positives detected, but stable and shuffled Lorenz-96 were false positives.
Nearest-neighbor divergence v1	Protocol failure	Stable fixture had no eligible positive-distance pair; no score was produced.
Nearest-neighbor divergence v2	Pass	Mackey-Glass 0.0196568 and Lorenz-96 0.0130529 promoted; stable/shuffled controls rejected or fail-closed as preregistered.

Observed. Nearest-neighbor divergence v2 was the first complete six-fixture Stage-A pass. **It was not a grammar or compression pass.** The fresh prefix-quantized Stage-B test then lost to LZ78 on both positives: Mackey-Glass 8,824.829 versus 744 bits; full-state Lorenz-96 8,482.142 versus 1,240 bits. This is the clearest decomposition in the project: geometric evidence can be present while the representation-plus-grammar code remains noncompetitive.

3.7 OmegaSim A6 integration

The first frozen OmegaSim detector run evaluated 180 rows: five seeds, couplings 0.35/0.60, delays 0/3, appraisal/linear/shuffled dynamics, and three feature strata. One of twelve preregistered cells promoted: coupling 0.60, delay 3, roles8, with four of five matched-seed wins. This was only bounded proxy evidence.

The subsequent stratified re-sweep completed all 120 planned role-specific A6 cells with three seeds and matched controls. Across the batches, every induced grammar had one or two productions and zero categories. Individual tuples occasionally achieved exploratory matched-control wins (including several 4/6, 5/6, and 6/6 cases), but none met the grammar-priority rule of more than two productions or at least one category.

Observed. The A6 sweep is complete under its frozen protocol. **Inferred.** It does not show that OmegaSim dynamics are simple; it shows that this CLA representation/search/coding pipeline does not expose nontrivial grammar in those traces. Repeating the same pipeline would mainly reproduce the same instrument limitation.

Provenance: `projects/omegasim/experiments/20260715T153049Z-cla-detector-preregistered` and batches ending at `20260718T084500Z-a6-stratified-cla-resweep-10`.

4 Failure decomposition

4.1 F1: high-dimensional symbolization

Evidence: 510 unique symbols in 1,024 points for 20D rectangular Lorenz-96; raw M1 perplexity around 258 versus 3.7-4.2 for low-cardinality controls.

Status: partially repaired. Train-fitted k-means/microstates restore recurrence. Linear VAMP did not outperform matched PCA/raw controls. The next representation should be justified by a held-out predictive or coding win, not by kinetic terminology alone.

4.2 F2: proposal/search coverage

Evidence: OmegaSim returns nearly invariant one- or two-production, zero-category grammars; a tiny oracle-like joint chunk/category move can improve an objective, but ordinary greedy results remain trivial. Synthetic performance varies sharply across source families.

Hypothesis. Greedy single-edit search is myopic: useful categories or hierarchical rules may require a temporarily costly coordinated move. Candidate mining may also fail to propose motifs whose occurrences vary in context or phase.

Missing diagnostic: For every iteration, record candidates proposed, support, data-bit saving, model-bit increment, total delta, overlap conflicts, and explicit rejection reason. Without this, “MDL penalty too high” and “search never found the grammar” are observationally confounded.

4.3 F3: model transmission cost

Evidence: held-out v1 spent roughly 32.5 kbits per model; indexed and compact codecs saved hundreds to thousands of bits and reversed some synthetic decisions, but hard positives still lost because their suffix improvement did not repay the model. Fibonacci and paperfolding are especially diagnostic: suffix coding improved substantially while total coding lost.

Hypothesis. The current grammar format pays for generality it does not use: symbol identifiers, rule tables, parse/index metadata, and categories may be encoded independently rather than through a source-adaptive universal prior. There may also be duplicated structural information between grammar and parse.

4.4 F4: data code conditional on the grammar

Evidence: on attractor fixtures, even 65,536-symbol fixed-model amortization remained behind Markov-1. This cannot be explained solely by a one-time model cost. The induced grammar’s conditional data code is itself too weak for these streams.

Hypothesis. Exact deterministic expansions are a poor fit to noisy/continuous dynamics. A probabilistic grammar or state-conditioned residual code may be needed: grammar for recurrent macrostructure, with a local Markov/arithmetic coder for deviations.

4.5 F5: objective mismatch

The algorithm seeks shortest exact code, while the scientific goal is compositional, intervention-relevant dynamical structure. LZ78 or Markov codes can win compression without yielding interpretable grammar, while a useful semantic grammar may initially cost more bits. Compression remains an essential anti-overfitting constraint, but one scalar objective may not be an adequate search guide.

Hypothesis. Use a two-stage or multi-objective scheme: first require held-out predictive gain beyond matched Markov/LZ controls, then select among competitive models using stability, category recovery, and intervention consistency.

4.6 F6: sample complexity and nonstationarity

The synthetic fixed-model source crossed only at 16,384 symbols. This confirms a real finite-sample threshold. But attractor runs failed even at 65,536 symbols, so “just use more data” is not a sufficient general explanation. Phase sensitivity and changing local partitions suggest that stationary symbol grammars may be mismatched to nonstationary orbit geometry.

5 Prioritized algorithm-repair programme

R0: Build a transparent score decomposition

Add one machine-readable row per candidate and iteration:

$$(\text{proposal type}, \text{support}, \Delta L_{\text{data}}, \Delta L_{\text{model}}, \Delta L_{\text{total}}, \text{accepted}, \text{reason}).$$

Also emit final totals by symbol dictionary, production topology, parameters, parse/index stream, and residual/literal escape stream.

Acceptance test: on the generalized complete-code fixture, the report must reproduce the observed 6,232-bit data saving, 6,296-bit model increment, and +64-bit rejection exactly.

R1: Oracle representability test

For each planted family, supply the true generating grammar in the current internal representation and compute:

1. oracle grammar plus suffix code;
2. best grammar found by current search;
3. external controls.

If the oracle loses, the representation/code is inadequate. If the oracle wins but search loses, the problem is mining/search. If both win but learned categories differ, the problem is identifiability rather than compression.

R2: Search ablations

On untouched synthetic families, compare:

- current greedy single move;
- beam search with widths 4, 16, and 64;
- paired chunk-plus-category moves;
- split/merge category moves;
- grammar initialization from suffix automata or Sequitur/Re-Pair;
- exact or branch-and-bound search on very small fixtures.

Match proposal sets and code accounting. Measure oracle recovery, total bits, runtime, and the first iteration at which the correct abstraction appears.

R3: Replace deterministic exact grammar with a hybrid code

Keep exact reconstruction, but encode:

$$L = L(G) + L(\text{macro parse} \mid G) + L(\text{residual symbols} \mid \text{state/context}).$$

Candidate residual coders should include KT/Dirichlet-smoothed Markov, context-tree weighting, and arithmetic coding. A grammar move should be credited only for the incremental conditional saving beyond the same residual coder without that move. This prevents grammar from being asked to explain every microscopic fluctuation.

R4: Use hierarchical and probabilistic categories

Current exact-context categories are brittle. Test soft/overlapping categories with a transmitted membership model, and hierarchical nonterminals with explicit reuse. Category success should be measured on planted latent classes: adjusted mutual information, held-out code gain, stability across seeds, and counterfactual substitution validity.

R5: Representation contest before grammar induction

Freeze a common downstream coder and compare:

- raw scalar/coordinate partitions;
- delay embeddings;
- PCA plus k-means;
- recurrence features;
- nearest-neighbor local geometry;
- nonlinear predictive embeddings;
- OmegaSim task-derived state factors.

Fit all transforms on the prefix only. Match alphabet size and occupancy. Select by held-out code/prediction, not by reconstruction variance. Linear VAMP should remain retired unless a new hypothesis predicts a specific advantage.

R6: Fresh calibration ladder

1. **Unit fixtures:** exact grammar recovery on tiny enumerable sources.
2. **Synthetic families:** substitution, hierarchical, stochastic, phase-shifted, and noisy grammars with exact-marginal nulls.
3. **Sample-complexity curves:** predeclare horizons and estimate the crossover distribution across seeds.
4. **Known dynamics:** fresh Mackey–Glass/Lorenz–96 trajectories, stable controls, and temporally structured surrogates.
5. **OmegaSim positive control:** implant a known recoverable grammar into an A6-like observation channel before testing endogenous dynamics.
6. **Untouched OmegaSim/A7:** only after the preceding gates pass.

The promotion rule should require every planted positive to beat literal, unigram, Markov-1/2, context-tree, and LZ78 controls; all nulls must be rejected; categories must recover planted structure where categories are part of the fixture.

6 Concrete first three experiments

ID	Question	Design	Decision rule
D1	Is failure representational, search, or coding?	Ten tiny planted grammars; enumerate or provide oracle grammar; compare oracle, greedy, beam, and external codes.	Proceed only if oracle beats controls on all positives and search recovers at least 8/10 without null promotion.

ID	Question	Design	Decision rule
D2	Is greedy myopia blocking categories?	Freeze proposal universe on complementary 3-uniform, Fibonacci, paperfolding, and two hierarchical noisy fixtures; compare greedy and paired/beam moves.	Search repair supported if it closes at least 80% of oracle-to-greedy bit gap on every fixture at bounded runtime.
D3	Can a hybrid probabilistic grammar code chaotic symbols?	Fresh prefix/suffix Mackey–Glass and full-state Lorenz–96; fixed representation contest; CTW/Markov residual under no-grammar and grammar models.	Both positives beat all complete controls; stable/shuffled controls not promoted; exact decode and frozen-prefix invariants pass.

7 What not to do

- Do not tune thresholds, partitions, horizons, or source parameters on the already inspected frozen suffixes.
- Do not treat the nearest-neighbor Stage-A pass as a CLA grammar pass.
- Do not replace complete-code comparisons with the older compression proxy.
- Do not infer “no chaos” from a grammar-coding null.
- Do not enlarge OmegaSim before a planted OmegaSim-like positive control is recoverable.
- Do not optimize only model serialization: the 65,536-symbol attractor failures show that conditional data coding and representation also matter.

8 Reproducibility map

Canonical project ledgers are under:

- `projects/chaos-language-algorithm/experiments/`
- `projects/omegasim/experiments/`

The most decision-relevant CLA ledgers are:

- 20260713T153839Z-lorenz63-rank-conditioned-vamp-tica
- 20260716T172310Z-frozen-heldout-calibration-v1
- 20260717T051500Z-generalized-complete-code-gate-v1
- 20260717T091500Z-indexed-complete-code-gate-v1
- 20260717T151800Z-frozen-heldout-calibration-v2
- 20260717T171800Z-synthetic-detector-calibration-v1
- 20260717T191500Z-synthetic-sample-complexity-v1
- 20260717T231900Z-synthetic-uniform-morphism-v1
- 20260718T071500Z-synthetic-compact-model-v1
- 20260718T111500Z-synthetic-amortization-v1
- 20260718T131500Z-attractor-amortization-v1
- 20260719T071500Z-vector-recurrence-attractor-v1

- 20260721T011500Z-nearest-neighbor-calibration-v2

The Stage-B prefix-quantized null is summarized in `projects/chaos-language-algorithm/DECISIONS.md` (July 21 decision) and `NOTES.md`; its frozen outcome remains binding. The complete A6 closure is recorded in `projects/omegasim/experiments/20260718T084500Z-a6-stratified-cla-resweep-10/RUN.md`.

9 Bottom line

Observed. CLA has crossed from an uncalibrated proxy prototype into a disciplined experimental system with exact codes, strong provenance, useful negative results, and one bounded geometric detector pass. It has not crossed the scientific gate that matters for OmegaSim: a nontrivial induced grammar that beats strong complete-code controls on fresh chaotic data and remains stable under matched nulls.

Inferred. The most likely repair is not one parameter change. The evidence supports a three-part redesign: transparent oracle-relative search diagnostics, a more expressive hierarchical/probabilistic grammar with a competitive residual coder, and representation selection by held-out coding rather than proxy compression. The proposed D1–D3 experiments can determine which part is decisive without another broad, ambiguous sweep.